

“After she fell asleep, it went to my next podcast, which was about a serial killer”: Unveiling Needs and Expectations Regarding Parental Control within Digital Assistant

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ABSTRACT

Voice-based digital assistants (for brevity, we term them as ‘digital assistants’ in this paper, unless otherwise specified) are getting increasingly popular in the household to automate digital routines. Hence, with communal usage of digital assistants in the household, children can gain access to content through digital assistants that are intended for adult audiences and parents are concerned about such risks. To this end, we created a low-fidelity prototype with designs of existing Google’s parental control for Google Assistant (*termed as baseline*), along with our features deriving from prior literature (*termed as treatment*). Then, we conducted a semi-structured interview with 7 parents to understand their perceptions, needs, and expectations surrounding parental control within digital assistants. As reflected in our results, digital assistants can play different roles in stimulating open communication between parents and children, augmenting parenting responsibilities to induce self-regulation and impulse control in children, and setting boundaries with granularity.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

voice-based digital assistants; google assistant; parental control; semi-structured interview; low-fidelity prototype

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1 INTRODUCTION AND RELATED WORK

As digital assistants make our digital routines automated, they have become an inevitable part of households in the U.S. These digital

assistants allow users to ask questions, plan the day, as well as facilitate control of smart home accessories with their voice. According to Pew Research Center [19], nearly half (46%) of Americans use digital assistants. Hence, digital assistants have been studied for many years in the realm of use and usability [18, 29], improved accessibility for people with physical impairments [10], use by special populations including families and children [4, 11, 13], people living with disabilities [23], and older adults [22], the personification of digital assistants [15, 22, 24], and privacy issues around the digital assistants [16].

With the proliferation of digital assistants in the household, children are also getting access to them and using them regularly. Prior studies have focused on understanding the family-centered usage of digital assistants [3, 25]. Beneteau et al. [4] explored how families learn about device capabilities together, and how they engage with, explore and discover new functionalities that support and extend daily family activities. A digital ethnography of home use of digital assistants conducted by Payy et al. [20] uncovers its benefits in improving children’s speech and asking them to complete the chores like taking a bath, notifying when it’s time to sleep or wake or when dinner was ready. Parents also wanted to teach children good manners, such as to say *thank-you* and *please*. Similarly, the study conducted by Beneteau et al. [3] reported how the use of Echo dot contributed towards fostering communication (*through speech and language practice and expanding communication skills*), disrupting access in the family (*resulting from undermining and interrupting each other’s use of Echo dot like giving contradictory instructions to it thereby needing to regulate the use*), and augmenting parenting (*as a third-party mediator, complementing parenting tasks, increasing children’s autonomy*). Amazon review analysis for parents’ perspectives of the Internet-connected toys revealed that parents wished for more parental controls and were concerned about jokes that were not age-appropriate [17].

However, there are very few studies to understand the needs and expectations regarding parental control from within digital assistants [12, 20]. To this end, we created a low-fidelity prototype incorporating the existing controls for Google Assistant in Google’s parental control app, Family Link, along with the features derived from prior literature and evaluated it through a semi-structured interview with 7 parents who have used some kind of personal digital assistant in their households. Through our study, we answer the following research question: **RQ1: What are the needs and expectations of parents regarding parental control from personal digital assistants?** Our findings from the study provide valuable insights to the CSCW community into the different roles of digital assistants



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in augmenting parenting responsibilities to induce impulse control in children, fostering open communication between parents and children, and setting boundaries with granularity.

2 METHODOLOGY

We conducted semi-structured interviews with 7 parents (see Table 1 for details) over Zoom to understand their perceptions regarding design concepts implemented in the form of a low-fidelity prototype using AdobeXD. The study took place between May and June 2022. Our study was approved by the Institutional Review Board (IRB) at our university.

Participant Recruitment. To recruit participants, we posted flyers on our university campus, and local neighborhoods. We shared recruitment messages in family housing groups on social media. We also used snowball sampling, recruiting a few participants from the recommendation of participants who had taken part in this study. To take part in our study, a participant has to be either a father or a mother of at least one child below 13 years of age, where the family has used some kind of personal digital assistant in their household and the child is an active user of it.

PID	Gender	Age Range	Children's Gender (Age)
P01	F	30-34	Male (12, 10, 7)
P02	F	30-34	Male (7), Female (10, 8, 4, 1)
P03	F	30-34	Male (10, 7)
P04	F	30-34	Male (6)
P05	F	40-44	Male (9, 7), Female (4)
P06	F	40-44	Male (10)
P07	F	35-39	Female (10, 8)

Table 1: Demographics of Participants (N=7)

Procedure. When a participant showed interest to take part in our study, we emailed them the Informed Consent Document (ICD). Once they agreed to the ICD, we scheduled a time for the online interview (audio recorded). At the start of the session, we asked a few questions about their experiences with parenting their children. Then, we shared a link with the participant over the Zoom chat box to go through the *baseline condition* (see Figure 1), and *treatment condition* (see Figure 2), where we gathered their feedback on the perceived usefulness of the features and their expectations. At the end of the interview, participants responded to a set of demographic questionnaires. On average, each session took between 20 and 30 minutes. The participants were compensated with a \$25 Amazon.com gift card for their participation.

Data Analysis. The interview audio recordings were transcribed and we performed thematic analysis [6–8] on the transcription of the audio recording, where we took an inductive approach. Each participant's data was coded by two independent researchers. They developed codes from the data of the first few participants, compared those codes, and then iterated again with more participants' data until we had developed a consistent codebook. Once the codebook was finalized, two researchers independently coded the remaining participants' data. After all participants' data had been coded, both researchers spot-checked the other's coded data and did not find any inconsistencies. Finally, we organized and taxonomized our codes into higher-level categories.

(a) Set content filtering for music

If you want Angel to listen to music on smart displays & speakers, you can choose services that won't let him play music that the services marked as explicit

[How music filters work](#)

- ☐ Play any music
Angel can listen to any available music without filters
- ☒ Play some music
Angel can listen to music from only these services that have filters
- ☒ YouTube - [Content settings](#)
- ☐ Spotify (non-explicit)
- ☐ Apple Music (non-explicit)
- ☐ Don't play music
Music is off limits for Angel

(b) Control for news & podcasts

Choose what happens when Angel asks Assistant for news & podcasts on smart displays & speakers. Angel might still be able to play news or podcasts in other media services you allow.

[How news & podcasts filters work](#)

- ☐ News
Choose what happens when Angel asks Assistant to play news. Some news might have mature content.
- ☐ News is allowed
- ☒ News is off limits
- ☐ Podcasts
Choose what happens when Angel asks Assistant to play a podcast. Some podcasts might have mature content.
- ☐ Podcasts are allowed
- ☒ Podcasts are off limits

(c) Control for assistant features

Assistant Features

- ☐ Calling
Choose whether Angel can make calls on smart displays & speakers
- ☐ Angel can only answer calls
- ☒ Angel can make & answer calls
- ☐ Answer to questions
Choose whether Assistant will answer basic questions like weather & time, or all questions, even if they're about mature topics
- ☐ Only answer basic questions
- ☒ Answer all questions
- ☐ Action for Families
Actions are like when you say "Hey Google, tell me a story." Approved developers can create more family-friendly Actions for Families
- ☐ Actions for Families are off limits
- ☒ Actions for Families are allowed

(d) Downtime for assistant

The smart displays & speakers you chose will stop responding to Angel, so he can take a break. He can still use alarms, timers, home controls and applications you choose.

Use schedule

- ☒ Days
School nights
- ☐ Time
8:00 PM - 6:00 AM

Figure 1: Baseline Condition

3 RESULTS

In this section, we present the findings from semi-structured interviews with our participants (see §2), which includes the participant's evaluation of our features in the treatment condition, along with the parental expectations from digital assistants. Following the guideline from prior research [1, 21, 26], for consistency, we use the following terms based on the frequency of themes in participants'

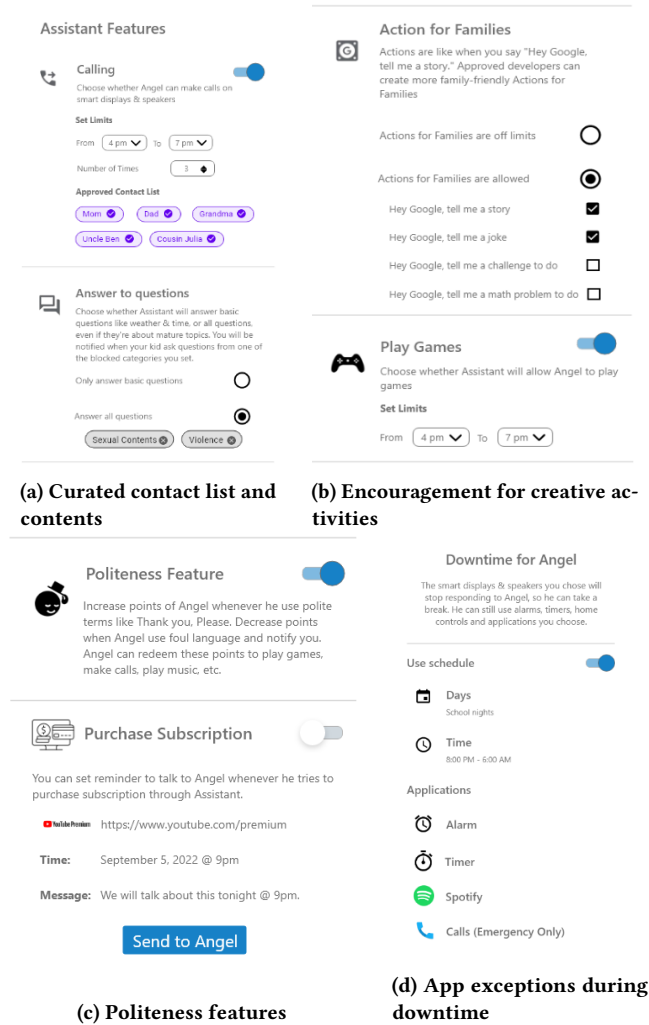


Figure 2: Treatment Condition

responses: a few (0-10%), several (10-25%), some (25-40%), about half (40-60%), most (60-80%), and almost all (80-100%).

3.1 Aide for Fostering Open Communication

Our participants expressed the need for having an open dialogue with their children to quench their curiosity regarding sensitive matters. In this regard, they expected the digital assistant to contribute towards the initiation of open communication from both parents and children.

3.1.1 Children's Initiative. Several mothers talked about the role of digital assistants in encouraging and prompting children to talk with their parents regarding any sensitive or mature topics. In this regard, one of the participants mentioned, *"Maybe like about mature matters, I'd want to be able to talk to them about and so I wouldn't want them going to a device, necessarily, besides just talking to a human about it. But at the same time, I don't want to discourage it...So yeah, I guess kind of having another option to say, ask any type*

of questions, but maybe certain types of content to block that, or to say, go back to your parents." (P07)

3.1.2 Parents' Initiative. On the other hand, some of the parents wanted to get notifications whenever children come up with inappropriate questions to the digital assistant so that they can explain it to their children as mentioned by one of the parents, *"I guess like if they asked an inappropriate question, I would like it to notify me in some way. So I know the things that he's wanting to learn about and things that we would need to talk about."* (P04)

3.2 Augmenting Parenting Responsibilities

Participants further highlighted the role of digital assistants as a neutral third-party mediator in complementing parenting responsibilities like regulating children's behavior and encouraging them to involve in creative activities.

3.2.1 Aide for cultivating good habits. With a view to regulating children's behavior and cultivating good habits in children, we introduced politeness features (see Figure 2c). Almost all of our participants found the feature to be interesting and wanted to see how this feature can dictate the real-life behavior of the children. Talking about the politeness feature, one of the parents expressed, *"The interesting thing is I've never seen it [politeness feature], it would be something that I would want to try and experiment with to see because they're basically trying to use it in a way to get more from it from the future. But if it gives them the habit of doing that in real life then I would be interested to see how it actually works."* (P07)

3.2.2 Encouragement for creative activities. Similarly, about half of the parents wanted digital assistants to encourage children to be involved in creative activities including solving challenges, telling stories, learning about science, and so on (see Figure 2b). They yearned to pivot children's usage of digital assistants toward productive and creative activities. For instance, one of the participants mentioned solving math problems with a digital assistant, *"It would be like, for instance, we're on summer break now. And it would be fun to have something like a math problem every day, that's a challenge or something, or, just giving kind of challenges that are more related to schoolwork activities, especially during the summer, and getting points for those kinds of challenges, being able to not just listen to a story, but maybe tell a story or finish a story or solve some type of a really fun feature to do."* (P01)

3.3 Setting Boundaries with Granularity

Our participants appreciated the features of granularity at different levels to attain a middle ground between appropriate usage and misuse of digital assistants by the children.

3.3.1 Curated contact list. Almost all of our participants admired the feature of the curated contact list (see Figure 2a), with whom the children can make contact through digital assistants. Most of them categorized grandparents, cousins, and parents themselves as the approved ones. In this regard, one of the participants explained how the curated contact list can aid parents in regulating children's usage of digital assistants, *"Yeah, so that's nice, just especially and I've had a few times, not necessarily all the time, but like, kids want to call someone too early in the morning."* (P01)

3.3.2 Filtering media contents by tags. Almost all of our participants expressed concerns about children accessing unintended content through digital assistants including swearing, violence, and sexual content, and parents wanted to filter those types of media content by tags (see Figure 2a). One of the parents expressed a similar concern, *“Swearing can’t happen and no sexual content can happen. We’ve never had really an incident with it. My kids don’t independently listen to news or podcasts yet. As they’ve gotten older, I’m pretty cautious about even listening to the news around them because I didn’t want them to casually hear about a school shooting. I’d rather have that conversation with them.”* (P05)

3.3.3 Exceptions for apps during downtime. While some of the parents valued downtime from digital assistants, they also expressed the need for some applications to be allowed during the downtime (see Figure 2d). Deriving from personal experiences of listening to bedtime stories with children on digital assistants, one of the participants mentioned, *“I think the only thing that I would maybe add is just specific apps like alarms and timers. I would maybe still want to allow Spotify at that time. I wouldn’t want them to have access to games or to ask a bunch of questions. But we do Spotify, like bedtime stories. They have a podcast that reads them, magic Treehouse, or whatever series they’re on, and they get an hour before bed. So maybe allowing some leniency for certain apps to do that.”* (P03)

3.4 Parental Expectations from Digital Assistant

Our participants also pointed to some of the expectations from digital assistants to streamline the perils of communal access of digital assistants.

3.4.1 Streamline perils of communal access. With communal access to digital assistants within families, there is a need for reliable distinction between kids and adults mode. Referring to one of the incidents with the digital assistant playing unintended podcasts, one of the parents expressed a similar need, *“So my daughter on the weekends, we do, like bedtime podcasts that she likes where they tell stories, but even that was in my menu. So like, after she fell asleep, it went to my next podcast, which was about a serial killer. Thank goodness she was asleep.”* (P05)

3.4.2 Access to interaction history. In order to regulate the use of digital assistants, several parents expected to have access to the interaction history of their children with digital assistants. One of the parents emphasized the need for accessing the interaction history, specifically in the context of suspicious activities, as quoted below, *“...if it were kind of record in that way, you could go back through the history and see which jokes or stories or challenges or whatever was being told. If they’ve got 15 minutes between when I’m doing something else, and they’ve been set up with an activity like that’s one that I haven’t noticed, actually comes up on the history. More just like flagged things, I think it could just be like flagged categories, flagged activity.”* (P03)

4 DISCUSSION

4.1 Roles of Digital Assistant

Our findings present the parents’ evaluation of features and expectations related to parental control from within the digital assistants. We found that digital assistants can play a crucial role in regulating children’s usage of devices through embedding positive values including inducing self-regulation, nurturing open communication, and setting boundaries through granularity.

Inducing self-regulation and impulse control in children. While prior work indicated the role of digital assistants in complementing parenting tasks including playing lullabies, telling stories, and helping with routines [3], our study uncovered the new contexts of augmenting parenting responsibilities through inducing self-regulation in children. For instance, almost all the parents from our study were interested to see how the politeness features can map the learning of children from digital assistants in terms of cultivating good habits to real-life behavior (see §3.2.1). Similarly, parents expected digital assistants to act as a neutral third-party mediators to encourage children to creative activities (see §3.2.2).

Nurturing open communication between parents and children. Prior literature in the field of parental mediation has found that fostering open communication can result in positive outcomes in children [5, 9, 28]. Our study has pointed to the role of the digital assistant in fostering open communication between both parents and children, whereby the parents expected the digital assistants to prompt children to ask their parents for sensitive matters as well as notify the parents so that they can initiate the conversation otherwise (see §3.1).

Attaining balance between use and misuse of devices. Several studies [14, 27] have suggested that an all-or-nothing type of control is less desired by parents, as coarsest controls hinder the development of children’s autonomy and learning opportunities, and weaken the mutual trust between parents and children. Hence, our study pointed towards how the principle of granularity at different levels can contribute to gaining a balance between appropriate usage and misuse of digital assistants within the family (see §3.3).

4.2 Limitations and Future Works

A limitation of our study is that we conducted semi-structured interviews with only seven participants, which followed a qualitative research approach [2, 6, 7]. We acknowledge a different set of samples might yield varying results. Thus, we do not draw any quantitative, generalizable conclusion from our study. Our studies were performed in a controlled environment, where the participants went through a low-fidelity prototype. We acknowledge the limitation of such studies that user response in a study environment may not be an exact representation of their behavior in the real world. In our study, we present the parent’s perceptions and expectations regarding parental control through digital assistants. Taking into account, children are one of the major stakeholders in the process, we plan to conduct a study with children to understand the suitability of the features in the context of their needs and expectations.

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